Workshop: The Modern ML Engineer - From Fine-Tuning to Deployment

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About me : Vijay Dwivedi

Current Engagement: AI/ML Consultant, TransPerfect/Lenovo Group

Profile Summary:

Over 20 years of expertise in **aviation, engineering**, **Machine Learning**, **Data Science & AI**.

Certified **Google Professional Machine Learning Engineer**

Experienced in Data Engg, Gen AI, and algorithm development

Key Competencies:

Proficient in **Python, SQL, TensorFlow and GCP**

Education:

M.Sc. (Mechanical), University of Strathclyde, Glasgow, UK

B.Tech (Mechanical), GB Pant University, Pantnagar, Uttarakhand, India

Certifications:

GE Analytics Engineer, Google Professional ML Engineer, etc

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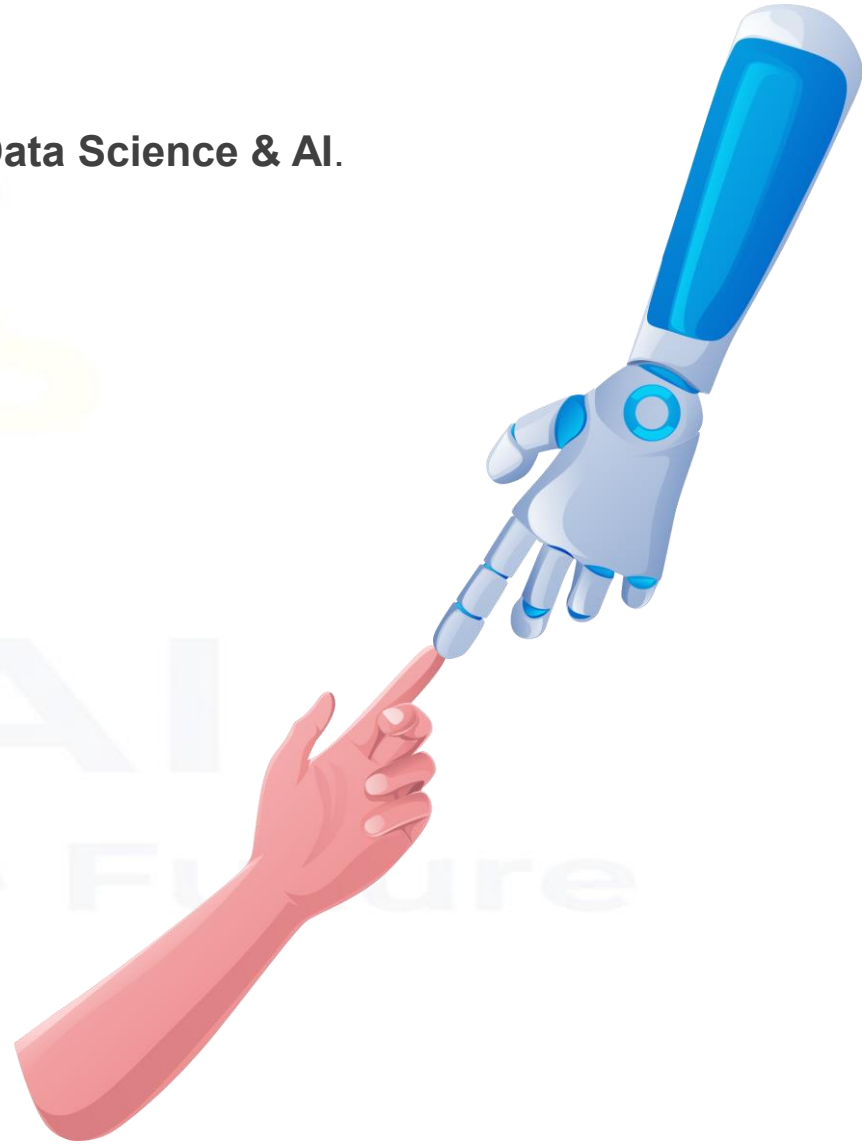

Our Guide for a Great Training Session

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Simple do’s and don’ts to help us maintain a smooth and productive training session.

Do:

- **Be Punctual:** Please join the meeting on time.
- **Be Present:** Mute your microphone when not speaking and turn on your video for better communication.
- **Active Participation:** Use the "Raise Hand" function or the chat box for questions and comments.

Don't:

- **Multitask:** Please give the session your full attention.
- **Minimize Noise:** Keep background noise to a minimum by staying muted.
- **Interrupt:** Be respectful and allow others to finish speaking.

Thank you for your cooperation in creating a focused and effective learning environment!

Workshop Group

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Workshop

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This workshop introduces to the modern machine learning engineering workflow, covering the complete lifecycle of a machine learning system from data preparation to deployment. Students will gain practical experience by building an end-to-end **Customer Support Intent Classifier** for an e-commerce use case.

Prerequisites: Basic knowledge of Python, Pandas, Scikit-learn, Transformer Models, Hugging Face, FastAPI, Docker and introductory machine learning concepts

Hour	Topic	Activity
1	Foundation & Data	Data cleaning and exploration
2	Feature Engineering	Embeddings and baseline model
3	Fine-Tuning	Transformer fine-tuning
4	API Development	Build prediction API
5	Deployment	Docker containerization
6	Industry Workflow	ML system overview

Use Case: Customer Support Intent Classifier for an e-commerce

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Hour 1 – Problem Framing and Data Preparation

- Students will be introduced to the problem of classifying customer support queries for an e-commerce system. They will set up their environment and explore a dataset containing raw customer messages such as return requests, refund inquiries, and product issues.
- **Activity**: Clean and prepare customer support queries and analyze intent categories.

Hour 2 – Feature Engineering and Baseline Model

- Students will convert customer queries into numerical representations using embeddings and train a baseline classifier to predict customer intent categories.
- **Activity:** Generate embeddings for customer queries and train a baseline intent classification model.

Hour 3 – Fine-Tuning a Transformer Model

- Students will fine-tune a pre-trained transformer model to improve classification performance on the customer support dataset.
- **Activity:** Fine-tune a transformer model to improve intent classification accuracy.

Hour 4 – Building a Machine Learning API

- Students will convert their trained intent classification model into a usable API that can accept customer queries and return predicted intents.
- **Activity**: Build an API endpoint that classifies customer support queries in real time.

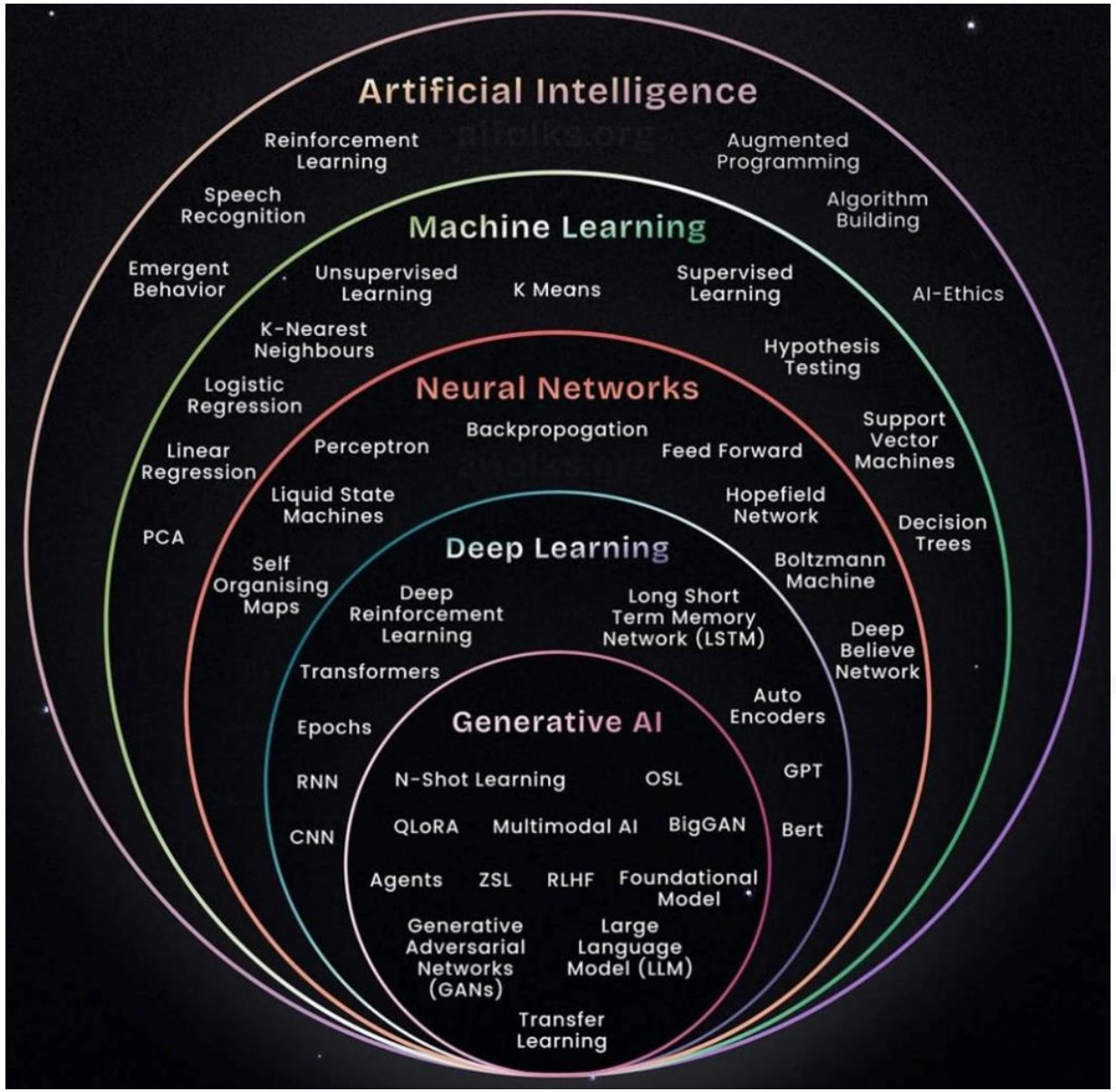
Hour 5 – Containerization and Production Concepts

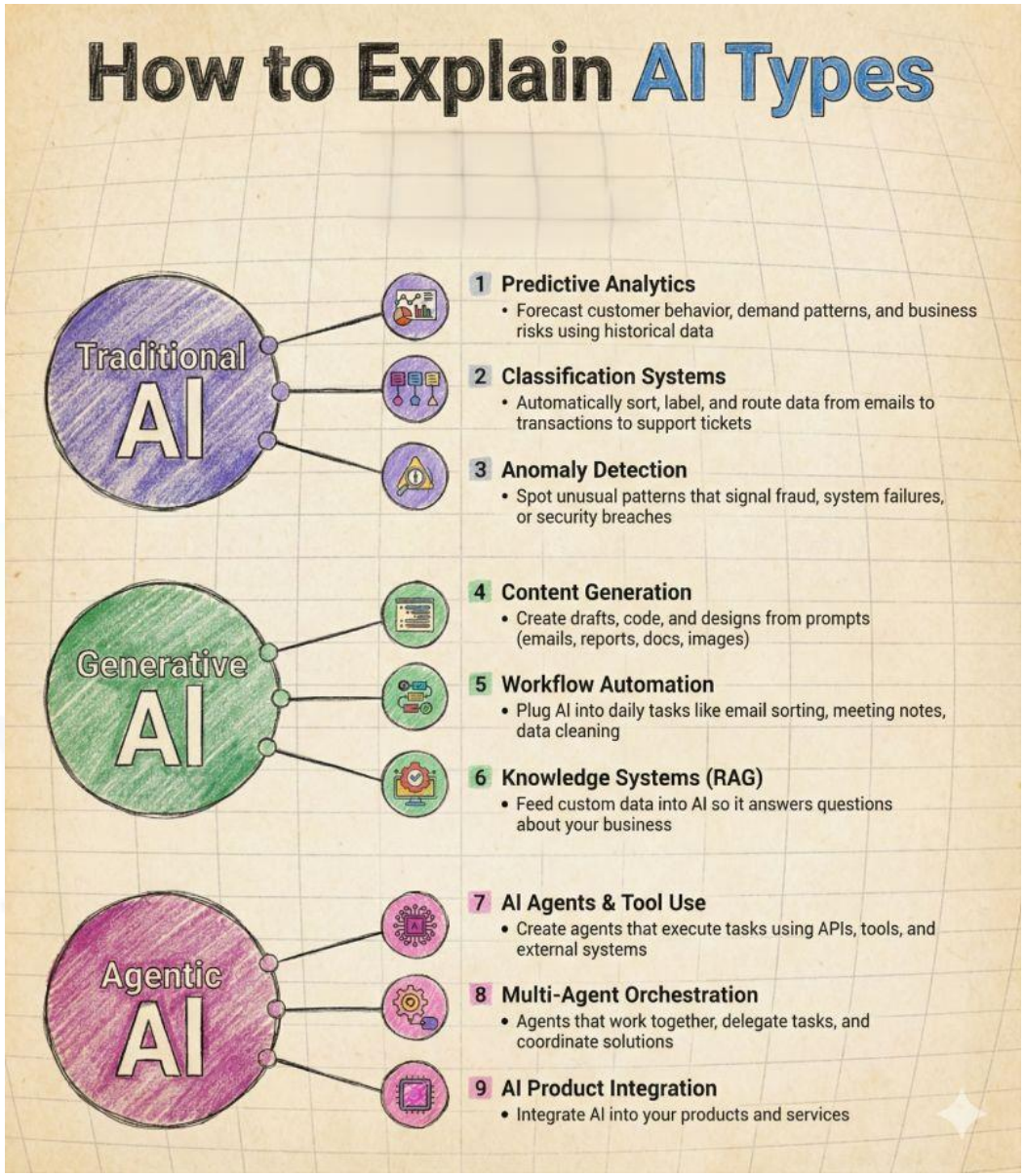
- Students will package their intent classification API into a container and learn how machine learning models are deployed in production environments.
  Activity: Containerize the intent classification API and run it locally

Hour 6 – Industry Workflow and MLOps Overview

- Students will learn how intent classification systems are deployed and maintained in real-world applications and how different roles contribute to ML systems.
- **Activity**: Understand how the intent classifier fits into a real-world ML deployment pipeline.

Artificial Intelligence





https://www.youtube.com/watch?v=Yq0QkCxoTHM

Large Language Models

- **Language Model:** Assigns probabilities to word sequences and generates text by sampling next words, trained on word counts from large text corpora (e.g., n-grams, HMM etc).

- **Large Language Model**: Similar goal but trained to predict the next word using massive datasets, learning broader language knowledge beyond word prediction. LLMs are AI systems used to model and process human language. They are called “large” because these types of models are normally made of hundreds of millions or even billions of parameters that define the model's behavior, which are pre-trained using a massive corpus of text data.

- The underlying technology of LLMs is called transformer neural network or **transformer** based on Google researchers in the famous paper **Attention is All You Need** in 2017.

- Transformers are capable of performing natural language (NLP) tasks with unprecedented accuracy and speed. With its unique capabilities, transformers have provided a significant leap in the capabilities of LLMs.

- LLM process flow: **Tokenization → Embedding → Attention via Transformer → Prediction → Response**

https://markov-lm.streamlit.app/

LLM : Step - 1 : Tokenization

- Text is broken into smaller units called tokens
  - 1 token = ¾ of words (rough approximation for large corpus)
- Tokens can be words, sub words, or characters
- Helps the model understand and process the input efficiently
- **“ Explain large language models to me like I am five years old”**

Explain large language models to me like I am five years old

https://platform.openai.com/tokenizer

LLM : Step - 2 : Embeddings

- Tokens are converted into numerical vectors
- These vectors represent the semantic meaning of words
- Embeddings enable mathematical operations and pattern recognition in text
- **“ Explain large language models to me like I am five years old”**
  - Explain = [0.2, 0.5, ……, 0.9]
  - Large = [0.45, 0.2, ……., 0.7]
- Similar words will take closure distance

http://projector.tensorflow.org/

LLM : Step - 3 : Transformer Mechanism

- The core of modern language models
- Uses self-attention to weigh importance of each word relative to others
- Helps the model focus on the most relevant parts of the input text
- **“ Explain large language models to me like I am five years old”**
- **“ large language model”** and **“ five years old”** are important group of words

https://poloclub.github.io/transformer-explainer/

LLM : Step - 4 : Prediction

- Based on context, the model predicts the next word or token
- Utilizes previous tokens and attention to ensure logical flow
- This is done for one token at a time until the sequence is complete
- **“ Imagine you have a super-smart robot friend who has read every ____.”**
- What is likely next word ? Book or Bank.

LLM : Step - 5 : Response Generation

- The predicted tokens are converted back into human-readable text
- The final result is a coherent and context-aware response
- This response is what the user sees
- **“ Imagine you have a super-smart robot friend who has read every book, story, and conversation ever written.”**
- When one asks a question, it looks like all the words it has learned and find the best answer, just like magic!

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Hour-2:Choosing the Right Error Measurement

You are asked to build a classifier for leukemia

- **Training data**: 1% patients with leukemia, 99%healthy
- **Measure accuracy**: total % of predictions that are correct
- Build a simple model that always predicts "healthy"
- Accuracy will be 99%...

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